Pranav Ghawat

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SUMMARY

Quantitative professional with **2+ years of experience** building **risk-management tools, pricing models and scenario-analysis frameworks** for fixed income and rates products, combined with hands-on trading experience in **US STIR and Treasury futures**. Strong grounding in **yield-curve relative value, macro-driven rate behavior and market liquidity dynamics**. Engineering graduate from **BITS Pilani**, skilled in Python, C++, Excel and **AI-assisted development workflows with coursework in derivative pricing**.

EXPERIENCE

Quantitative Researcher | *Definedge Securities* January 2026 – Present

- **Options trading tools:** Developed a multi-leg **options P&L and risk pricer** incorporating **Delta, Gamma, Theta and Vega**, enabling analysis of spreads, flies and other user-defined strategies.
- **Derivatives analytics:** Led development of Rzone's **options strategy framework**, supporting same-expiry and calendar spreads/flies and translating relative-value concepts into retail trading tools.
- **Quantitative trading tools:** Built **regression, correlation and pairs-trading scanners** in Python/C++, including cointegration and optimal hedge-ratio analysis; developed a strategy **backtesting engine**, driving 30–40% user adoption.
- **AI-accelerated development:** Used **GenAI tools** (e.g., Claude, Copilot) for code generation, debugging and workflow automation across the build cycle, applying manual review to ensure correctness before deployment.

Market Analyst | *Futures First Info Services Pvt Ltd* January 2024 – May 2025

- **Rates trading:** Traded **US STIR and Treasury futures across 20+ maturities** on CME/CBOT, running **relative-value strategies** (spreads, butterflies, condors, ratio trades) to capture curve mispricings, alongside outright and basis trading of the **SOFR–Fed Funds money-market spread**, with full ownership of execution, risk and P&L.
- **Macro-economic understanding:** Built a strong working model of **yield-curve shape formation** – how Fed policy path, inflation surprises (CPI/PCE), labor data (NFP) and growth expectations drive curve steepening, flattening and inversion – used to anticipate curve repositioning ahead of key releases.
- **Market microstructure:** Traded live through major liquidity-shock events, including the **August 2024 yen carry-trade unwind** and the **April 2025 tariff-shock volatility**, directly observing liquidity air-pockets, spread widening and order-book fragility under stress.
- **Quantitative tooling:** Sole analyst in the batch to build a **Python pipeline** to clean and structure historical SOFR and Fed Funds futures data for historical curve construction and analysis; also built **Excel-based pricing models** and used Bloomberg Terminal/Refinitiv Eikon for real-time monitoring.
- **Risk management:** Tracked **50+ macro events monthly** (FOMC, CPI, NFP, central bank meetings), running Monte Carlo simulations, stress tests and scenario analysis to model rate-path outcomes and P&L impact across the curve.
- **Execution automation:** Built a proprietary **ADL-based execution algorithm** and automated multi-leg execution using Trading Technologies' Autospreader, achieving **70% fills at target price with no payup**.
- **Performance – Top of batch:** Ranked **#1 in the 24-member analyst cohort** on trading volume, averaging **15,000 round turns/month** with **90% directional accuracy** on yield-curve trade calls, while scaling book size responsibly with disciplined VaR limits (up to $80K).
- **Recognition:** Selected **1 of 4 analysts** from the 24-member cohort for a **Pre-Placement Offer**; recognized by desk VP, manager and Branch Head for consistent trade quality.

Quantitative Analyst Intern | *Futures First Info Services Pvt Ltd* July 2023 – December 2023

- **Quantitative research:** Built and calibrated **Vasicek, CIR and Hull-White** short-rate models using 60+ months of US Treasury and SOFR futures data; applied maximum likelihood estimation and numerical optimization.
- **Curve construction & backtesting:** Constructed yield curves using **Nelson-Siegel and cubic-spline interpolation** and built a vectorized Python backtesting framework for curve and relative-value strategies, reducing research runtime by **80%**.

EDUCATION

B.E. (Hons), Chemical Engineering 2024

Birla Institute of Technology and Science (BITS), Pilani *8.01/10*

Relevant Coursework: Derivatives and Risk Management, Business Analysis and Valuation, Fundamentals of Finance and Accounting, Numerical Methods

SKILLS

Markets & Trading: Fixed Income, US Rates/STIR, Futures, Options, Yield-Curve Modelling, Relative Value, Risk Management (VaR)

Analytics: Pricing, P&L Analysis, Monte Carlo/Scenario Analysis, Financial Modelling, Backtesting

Programming & Tools: Python, C++, C, R, MATLAB, Excel, Bloomberg Terminal, Refinitiv Eikon, Trading Technologies Autospreader

AI Tools: GenAI-assisted development (code generation, debugging, workflow automation)

Languages: English, Marathi, Hindi